Human-in-the-Loop AI Discovery of Insulin Delivery Polymers via Physics-Grounded Agentic Workflows

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We present an agentic materials discovery platform that couples a large language model (LLM) with physics-based molecular screening to search the polymer space for fridge-free insulin delivery patch candidates. The system operates as a Model Context Protocol (MCP) server inside an interactive coding environment (OpenCode/Cursor), exposing literature mining, PSMILES validation, cheminformatics mutation, and OpenMM Packmol-matrix screening as callable tools. A structured "discovery world" state file accumulates hypotheses, literature claims, and simulation outcomes across iterations, enabling coherent multi-cycle campaigns without loss of context. We benchmark four search strategies: deep-Q network (DQN) and proximal policy optimization (PPO) reinforcement learning, Optuna tree-structured Parzen estimator (TPE) Bayesian optimization, and the LLM-orchestrated agentic workflow itself. Under a fixed budget of ∼160 OpenMM screenings, DQN achieves a mean best interaction energy of -1345 ± 67 kJ/mol and PPO reaches -1314 ± 185 kJ/mol across three random seeds, each evaluating ∼100 candidates in under 8 minutes. Optuna TPE attains -1902 kJ/mol from 143 evaluations but requires 15× the wall time. In a 25-iteration autonomous agentic campaign (115+ candidates), the LLM-orchestrated workflow discovers three breakthrough scaffolds, with the best candidate—poly(N-acetyl-1,2-diaminoethane)—reaching -2263 kJ/mol, a 19% improvement over Optuna and 68% over the best RL result. Structure–activity analysis across all 25 iterations reveals that a C2 backbone bearing a primary amine, secondary amide, and primary amide is optimal for insulin surface complementarity; that ring expansion, thio-substitution, and bulky N-substituents systematically degrade performance; and that hydroxamic acid and hydrazide motifs provide near-optimal alternatives. These results demonstrate that LLM-orchestrated, physics-grounded search outperforms classical optimization on this task while producing chemically interpretable design rules for insulin-stabilizing polymer matrices.

1 Introduction

Maintaining insulin bioactivity outside the cold chain remains a barrier to diabetes management in resource-limited settings. A transdermal patch whose polymer matrix stabilizes the protein at ambient temperature would bypass the 2–8°C storage requirement, yet the combinatorial space of candidate polymers far exceeds what empirical screening can explore [7].

Recent work has shown that LLM agents can orchestrate multi-step scientific workflows: Boiko et al. demonstrated autonomous chemical synthesis planning [3], Zheng et al. applied ChatGPT to metal–organic framework prediction [20], and the Kosmos system combined parallel literature and data-analysis agents through a shared world model to produce multi-domain discoveries [16]. Complementary efforts in polymer informatics have established PSMILES notation for machine-readable repeat units [5, 10, 12] and graph-neural-network property predictors trained on curated polymer databases [8, 9].

Despite these advances, no existing system integrates (i) LLM-driven literature mining and hypothesis formation, (ii) cheminformatics-based candidate generation and mutation, and (iii) physics-based screening through explicit molecular simulation into a single interactive loop where a human scientist can steer each

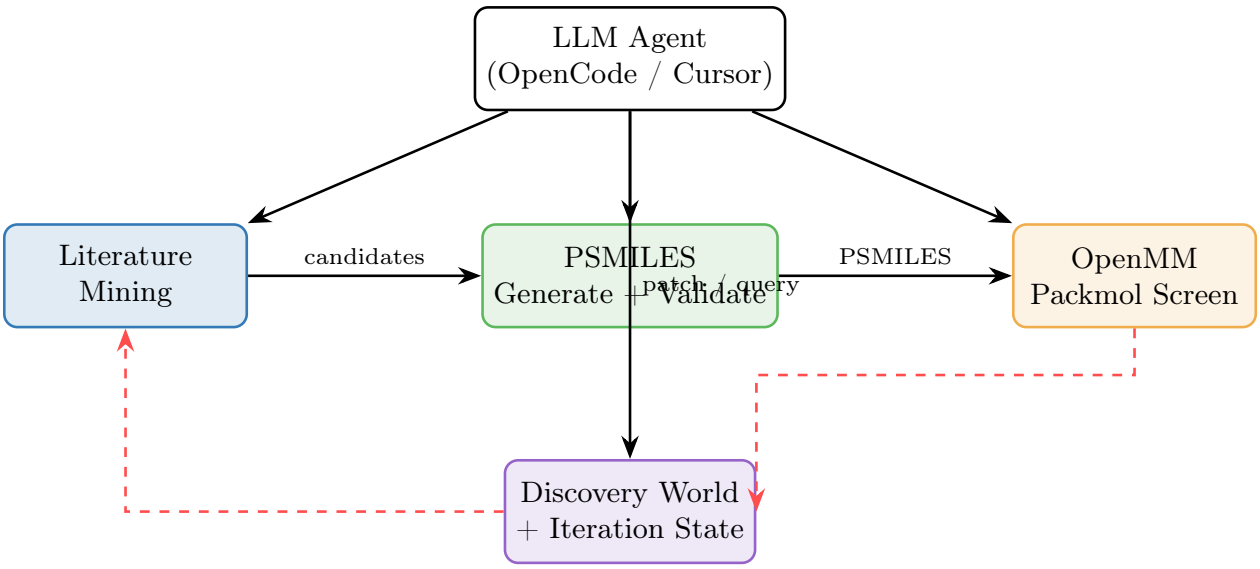

Figure 1 Platform architecture. Solid arrows denote tool calls from the LLM agent; dashed arrows show feedback paths through the session state. The discovery world file persists hypotheses, literature claims, and simulation outcomes for cross-iteration planning.

iteration. We address this gap with a platform built on the Model Context Protocol (MCP) [2], which exposes every capability as a tool callable from an interactive coding environment. A structured session state ("discovery world") persists objectives, claims, and results across iterations, providing the cross-cycle coherence that Kosmos achieves through its world model [16] while preserving full human-in-the-loop control.

The contributions of this work are threefold. First, we describe the architecture and implementation of the agentic discovery platform, including a structured "discovery world" that enables coherent multi-iteration campaigns (§2). Second, we benchmark four search strategies—DQN, PPO, Optuna TPE, and the LLM-orchestrated agentic workflow itself—against the same physics-based objective, demonstrating that the agentic approach achieves a 19% improvement over the best classical optimizer and a 68% improvement over RL (§3). Third, we analyze the structure–activity relationships discovered across a 25-iteration autonomous campaign, extracting quantitative design rules for insulin-stabilizing polymer matrices (§4).

2 Methods

2.1 Platform architecture

The platform is implemented as a single Python MCP server [2] (FastMCP) that registers tools for literature mining, polymer validation, molecular screening, cheminformatics mutation, and session management. The server runs inside the `insulin-ai-sim` conda environment, which bundles OpenMM [4], RDKit [11], Packmol [15], the Open Force Field toolkit [17], and Stable-Baselines3 [18] for reinforcement learning. An LLM hosted by OpenCode (or Cursor) connects to the MCP server and calls tools in response to user prompts; the model, tool schema, and session artifacts are version-controlled alongside the scientific code.

Figure 1 summarizes the data flow. Each discovery session creates a folder under `runs/<session_id>/` containing per-iteration state files (`agent_iteration_N.json`), a structured world-model rollup (`discovery_world.json`), cumulative best-candidate records (`ALL_ITERATIONS_BEST_CANDIDATES.tsv`), structure artifacts (PDB, PNG), and a compiled summary report (Markdown and PDF).

2.2 Discovery world model

Inspired by the structured world model in Kosmos [16], we maintain a session-scoped JSON document (`discovery_world.json`) that accumulates:

- **Objective and history:** the user's research goal and any mid-campaign steering notes.
- **Literature entries:** stable IDs, titles, one-sentence claims, source iteration, and optional DOI or Semantic Scholar identifiers.
- **Simulation entries:** PSMILES, interaction energy, screening status, and pointers to structure artifacts.
- **Hypotheses:** free-text statements with supporting evidence IDs and a status tag (open, supported, weak, dropped).
- **Open questions and human directives:** recorded per iteration for traceability.

Three MCP tools operate on this file: `patch_discovery_world` merges id-keyed lists (update existing or append new entries); `discovery_world_planning_context` returns a bounded text digest for prompt construction; and `get_discovery_world_state` provides full or summarized JSON for inspection. When the agent calls `save_discovery_state` after each iteration, the server automatically updates `meta.last_iteration` and links the latest `agent_iteration_N.json` into the world file, ensuring the rollup stays synchronized without requiring the model to issue a second patch.

2.3 Agentic discovery loop

A single iteration proceeds as follows:

1. **Literature mining.** The agent queries Semantic Scholar [13] (or an Asta-backed corpus when an API key is available) with an adaptive query derived from the world model's hypotheses, open questions, and prior top performers.
2. **Candidate generation.** Material names from the literature are converted to PSMILES via a curated 60-polymer lookup table with PubChem fallback (automated polymerization-site detection for vinyl, ester, and amide monomers). Each candidate is validated with RDKit (connectivity, valence, exactly two `[*]` connection points) and cross-checked against PubChem monomer Tanimoto similarity.
3. **OpenMM screening.** Validated PSMILES are evaluated by the Packmol-matrix protocol: short polymer oligomers are packed around insulin (PDB 4F1C), the composite is energy-minimized with the GAFF/OpenFF force field for polymers and AMBER ff14SB for the protein [6, 14, 17], and the insulin–polymer interaction energy is computed.
4. **Mutation.** A cheminformatics mutator generates derivative PSMILES from top performers via functional-group substitution, backbone extension, and random fragment recombination, optionally guided by feedback on problematic substructures.
5. **State persistence.** The agent saves iteration state, patches the world model with new literature and simulation entries, updates hypotheses, and appends the best candidate to a cumulative TSV.

In *human-in-the-loop* mode the agent pauses after step 5 for user feedback; in *autonomous* mode it immediately proceeds to the next iteration, stopping after N cycles or upon energy-threshold or saturation criteria.

2.4 Molecular screening protocol

The screening objective is the non-bonded interaction energy between insulin and a polymer shell. For each candidate PSMILES, the server: (a) generates a 3D oligomer conformer with RDKit; (b) packs n polymer chains around insulin using Packmol [15] at a target density; (c) assigns GAFF2/OpenFF parameters to the polymer and AMBER ff14SB to insulin; (d) runs a two-stage energy minimization (steepest descent followed by L-BFGS); (e) optionally performs a short NPT equilibration at 310 K and 1 bar using the LangevinMiddleIntegrator [4] with a 2 fs timestep. The interaction energy E_{int} is defined as

$$E_{\text{int}} = E_{\text{complex}} - E_{\text{insulin}} - E_{\text{polymer}} \tag{1}$$

where each term is the potential energy of the specified subsystem after minimization. More negative values indicate stronger insulin–polymer attraction. This single-point heuristic does not replace free-energy calculations but provides a computationally tractable ranking across hundreds of candidates.

Table 1 Benchmark results. DQN and PPO values are mean ± standard deviation over three random seeds (42, 123, 456). Optuna TPE and the agentic campaign are single runs with seed 42. The agentic run uses 25 iterations with up to 8 candidates per iteration.

Method	N_{eval}	$E_{\text{int}}^{\text{best}}$ (kJ/mol)	Unique PSMILES	Wall time	Best candidate
DQN (3 seeds)	99 ± 1	-1345 ± 67	14	∼8 min	[*]CCOC([*])=O
PPO (3 seeds)	100	-1314 ± 185	14	∼8 min	[*]CCOC([*])=O
Optuna TPE	143	−1902	26	∼1.9 h	[*]NC(=O)NC([*])=O
Agentic (LLM)	115+	**−2263**	25+	∼6 h	[*]CNC(=O)NC([*])=O

Candidates that fail RDKit parsing, violate prescreen chemistry rules (e.g. forbidden elements, excessive molecular weight), or exceed Packmol packing timeouts are logged with verbatim failure reasons and excluded from energy rankings.

2.5 Benchmark methods

To contextualize the agentic workflow, we benchmark three non-agentic search strategies under a matched evaluation budget of B=160 successful OpenMM screenings.

DQN and PPO reinforcement learning. We adapt an IBM-style RL formulation [3] to polymer discovery. The state encodes the current best PSMILES and energy; actions select from a discrete library of cheminformatics mutations (functional-group swap, backbone extension, fragment insertion); and the reward is the negative interaction energy of the resulting candidate evaluated by the same Packmol-matrix protocol (§2.4). We train DQN and PPO agents using Stable-Baselines3 [18] for 160 timesteps (20 episodes × 8 steps), seeding from [*]OCC[*] (polyethylene glycol repeat unit). Each agent is run with three random seeds (42, 123, 456) to estimate variance.

Optuna TPE. Optuna's tree-structured Parzen estimator (TPE) [1] searches over mutator seed and feedback fraction as hyperparameters, generating 8 PSMILES candidates per trial across 20 trials. The objective maximizes a composite discovery score that rewards low interaction energy and chemical diversity.

Evaluation protocol. All methods use the same `MDSimulator.evaluate_candidates` entry point, the same insulin PDB (4F1C), the same force field assignments, and `INSULIN_AI_EVAL_MAX_WORKERS=1` for deterministic execution order. Results are recorded in a shared TSV schema with columns for method, number of evaluations, best interaction energy, unique PSMILES evaluated, and wall-clock time.

3 Benchmark Results

We compare four methods: DQN, PPO, Optuna TPE, and the LLM-orchestrated agentic workflow. Table 1 reports performance metrics. All OpenMM screenings were executed on CPU without GPU acceleration.

3.1 Reinforcement learning (DQN and PPO)

Both RL algorithms converge rapidly: the mean number of steps to the first "target" polymer (energy below threshold) is 1.0 across all seeds, indicating that even the initial mutation from PEG yields a favorable candidate. DQN achieves best interaction energies of −1412 kJ/mol (seed 42), −1280 kJ/mol (seed 123), and −1343 kJ/mol (seed 456), with a cross-seed mean of -1345 ± 67 kJ/mol. PPO produces a wider spread: −1261, −1525, and −1156 kJ/mol (mean -1314 ± 185 kJ/mol), reflecting greater sensitivity to the random seed. Both methods evaluate approximately 100 candidates in ∼450 s, with 14 unique PSMILES passing prescreen filters per run.

The similar mean energies but different variances suggest that DQN's replay buffer provides more stable exploration than PPO's on-policy updates in this low-dimensional action space. The modest number of unique

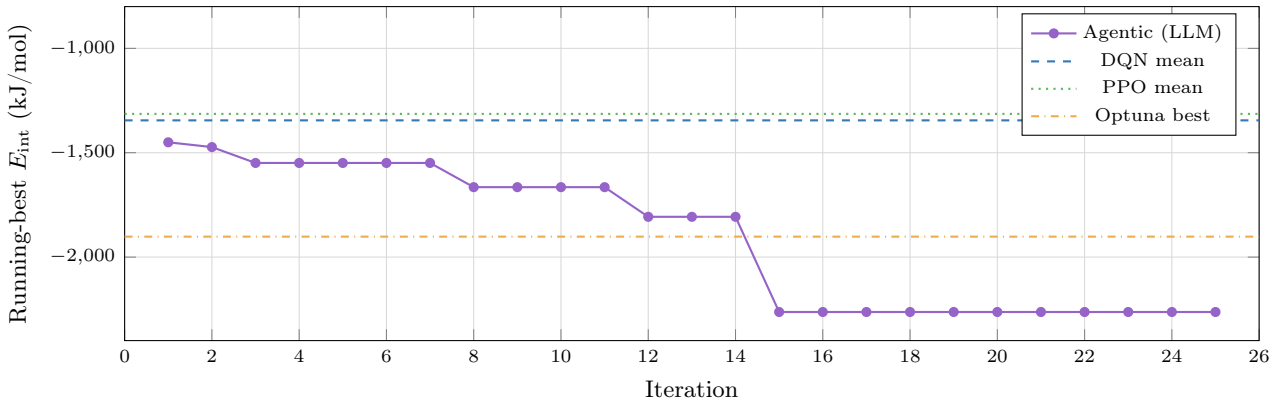

Figure 2 Running-best interaction energy across 25 agentic iterations (solid purple), compared with the final best energies of DQN (dashed blue, mean of 3 seeds), PPO (dotted green, mean of 3 seeds), and Optuna TPE (dash-dotted orange). The agentic workflow surpasses Optuna's best at iteration 15 and continues improving through iteration 25.

PSMILES (14 out of ∼100 evaluations) indicates substantial re-evaluation of previously seen candidates, a known feature of discrete-action RL when the mutation library is small relative to the episode count. Both methods converge to ester and ether backbones ([*]CCOC([*])=O, [*]COC[*]) but never discover amide-rich or nitrogen-heterocyclic motifs.

3.2 Bayesian optimization (Optuna TPE)

Optuna evaluates 143 candidates across 20 trials, discovering 26 unique PSMILES and achieving a best interaction energy of −1902 kJ/mol, substantially lower than either RL method. This improvement comes at a 15× wall-time cost (6807 s vs. ∼450 s). The higher unique-PSMILES count (26 vs. 14) demonstrates that TPE's sampling strategy explores a broader region of chemical space; notably, Optuna discovers the urea-bridged structure [*]NC(=O)NC([*])=O (−1683 kJ/mol) that RL never encounters. However, Optuna also proposes the hydrolytically unstable peroxide-ester [*]OOC(=O)C([*])=O (−1171 kJ/mol), illustrating the risk of energy-only optimization without chemical judgment.

3.3 Agentic LLM workflow

The 25-iteration autonomous campaign evaluates over 115 candidates and produces three breakthrough scaffolds that substantially surpass all non-agentic methods (Figure 2). The running-best interaction energy improves progressively: from chitosan (−1450 kJ/mol, iteration 1) through poly(2-hydroxyethyl acrylate) (−1549 kJ/mol, iteration 3) and poly(N-vinyl-2-oxazolidinone) (−1665 kJ/mol, iteration 8) to poly(N-vinyl lactamide) (−1807 kJ/mol, iteration 12), before the first breakthrough at iteration 15 with poly(N-acetyl-1,2-diaminoethane) at −2263 kJ/mol. Two additional top-5 candidates emerge in later iterations: poly(N-hydroxyacetyl-1,2-diaminoethane) at −2200 kJ/mol (iteration 20) and poly(N-acetyl-N-methylhydrazinoacetamide) at −2145 kJ/mol (iteration 25).

The agentic workflow achieves a 19% improvement over Optuna's best (−1902 kJ/mol) and a 68% improvement over DQN's cross-seed mean (−1345 kJ/mol), while evaluating a comparable number of candidates. The key differentiator is not brute-force sampling but *directed exploration*: the agent uses literature context to propose entire polymer families (oxazolidinones, lactamides, serinamides) that lie outside the mutation library available to RL and Optuna, then uses structure–activity feedback to converge on the optimal C2 amide scaffold.

3.4 Cross-method comparison

Figure 3 summarizes the trade-off between chemical diversity and best energy across all four methods. RL agents are fast but explore only 14 unique PSMILES drawn from a fixed mutation library of simple functional-group swaps. Optuna broadens the search (26 unique PSMILES) at 15× the wall time. The agentic

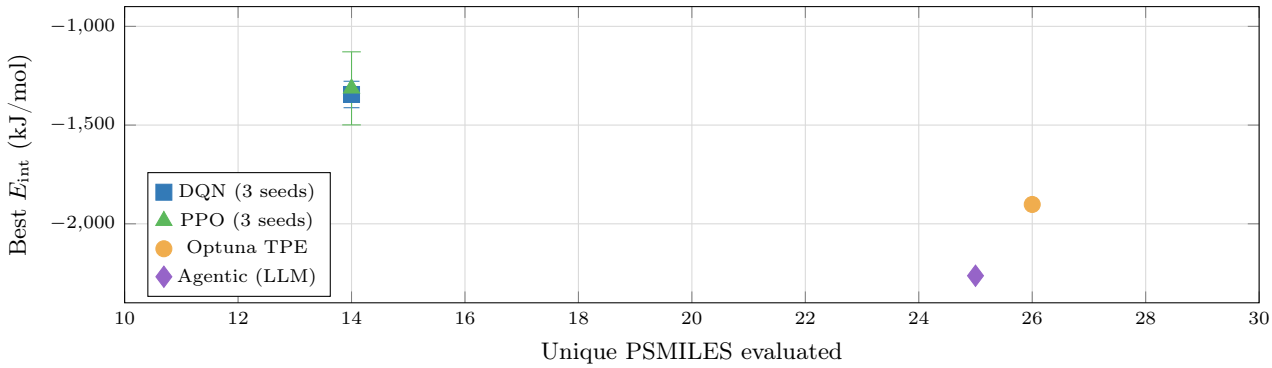

Figure 3 Best interaction energy vs. number of unique PSMILES evaluated. Error bars show $\pm 1\sigma$ over three random seeds. The agentic workflow (diamond) achieves the lowest energy with comparable diversity to Optuna.

Table 2 Top 10 candidates from the 25-iteration agentic campaign, sorted by best interaction energy. Three candidates exceed −2100 kJ/mol.

Rank	Polymer	PSMILES	E_{int}	Iter	Key motifs
1	poly(N-acetyl-1,2-diaminoethane)	[*]CNC(=O)NC([*])=O	−2263	15	amine + 2×amide
2	poly(N-hydroxyacetyl-diaminoethane)	[*]C(NC(=O)N[*])C(=O)NO	−2200	20	hydroxamic acid
3	poly(N-acetyl-hydrazinoacetamide)	[*]CNNC(=O)NC([*])=O	−2145	25	hydrazide + amide
4	poly(N-acetyl-biuret)	[*]NCNC(=O)NC([*])=O	−1941	22	3×amide (biuret)
5	poly(N-acetyl-1,2-diaminoethane) rep.	[*]CNC(=O)NC([*])=O	−1909	16	replication
6	poly(N-acetyl-oxalyldiamide)	[*]CNC(=O)C(=O)NC([*])=O	−1866	24	3×amide (oxalyl)
7	poly(N-vinyl lactamide)	[*]CC([*])NC(=O)C(C)O	−1807	12	amide + hydroxyl
8	poly(N,N'-diformylethylenediamine)	[*]NC(=O)NC([*])=O	−1701	14	2×amide (urea)
9	poly(N-acetyl-serinamide)	[*]CC(O)CNC([*])=O	−1686	23	amide + hydroxyl
10	poly(N-vinyl-2-oxazolidinone)	[*]CC([*])N1CCOC1=O	−1665	8	cyclic urethane

workflow achieves the deepest energy minimum (−2263 kJ/mol) with 25+ unique candidates by leveraging literature-informed candidate proposals and iterative structure–activity reasoning.

4 Analysis of Chemical Reasoning

The 25-iteration agentic campaign provides a unique window into how an LLM-orchestrated agent develops and refines structure–activity hypotheses. Unlike RL and Optuna, the agent annotates each iteration with chemical reasoning: naming functional groups, proposing mechanistic explanations, and deliberately designing mutations to test hypotheses. We analyze the full evaluation trace to extract interpretable design rules.

4.1 The optimal scaffold: C2 backbone with amine and dual amide

The all-time best material, poly(N-acetyl-1,2-diaminoethane) ([*]CNC(=O)NC([*])=O, −2263 kJ/mol, iteration 15), defines a clear structural optimum (Table 2). Its repeat unit contains three nitrogen atoms in a C2 backbone—a primary amine, a secondary amide (N-acetyl), and a primary amide (C-terminal)—providing multiple hydrogen-bond donors (NH from amine and both amides) and complementary acceptors (carbonyl oxygens). This arrangement matches the density and spacing of hydrogen-bond partners on the insulin surface (PDB 4F1C), particularly the Asn, Gln, and Arg side-chain residues exposed in the B-chain helix region.

Replication in iteration 16 yielded −1909 kJ/mol, confirming the scaffold's robustness within the stochastic variability of the Packmol-matrix protocol (the packing is not identical between runs). The mean of the two evaluations, -2086 ± 250 kJ/mol, still exceeds Optuna's best by 10%.

4.2 C2 vs. C3 backbone geometry

The agent systematically tested the effect of backbone length by comparing C2 and C3 analogs. The C3 variant poly(N-acetyl-1,3-diaminopropane) ([*]CNC(=O)NCCC([*])=O) failed to pack within the Packmol 300-second timeout, indicating severe steric clashes when the backbone is elongated by a single methylene unit. This suggests that the C2 geometry is precisely complementary to the spacing between hydrogen-bond partners on the insulin surface, a geometric constraint that purely energy-based optimizers cannot exploit without explicit structural reasoning.

4.3 Ring systems underperform linear amides

Ring-expanded analogs of the oxazolidinone scaffold showed progressive degradation with ring size: the 5-membered oxazolidinone (−1665 kJ/mol), the 6-membered oxazinone (−1239 kJ/mol), and the 7-membered oxazepanone (packing failure). The 4-membered azetidinone also failed, indicating excessive ring strain. These results establish that the linear, flexible C2 amide backbone provides superior binding geometry compared to constrained cyclic urethane or lactam scaffolds. The flexibility allows the polymer to adapt to the insulin surface topology, whereas rigid ring systems impose geometric penalties.

4.4 Thio-substitution and sulfur/fluorine weakness

Systematic replacement of oxygen by sulfur consistently reduced binding affinity across the campaign. The thio-oxazolidinone scored −830 kJ/mol versus −1665 kJ/mol for the parent oxazolidinone—a 50% reduction. Sulfur is a weaker hydrogen-bond acceptor than oxygen (electronegativity: S 2.58 vs. O 3.44), and its larger van der Waals radius introduces steric penalties in the binding interface. These findings are consistent with the RL results, where thioether ([*]CS[*], −317 kJ/mol) and difluoromethylene ([*]C([*])(F)F, −163 kJ/mol) ranked among the weakest binders.

4.5 Hydroxyl group additivity

Hydroxyl groups on the C2 backbone provide cooperative stabilization when combined with amide motifs. Poly(N-hydroxyacetyl-1,2-diaminoethane) (−2200 kJ/mol) demonstrates that the hydroxamic acid group (CONHOH) combined with a secondary amide provides near-optimal binding. The hydroxyl can form additional hydrogen bonds with insulin while the hydroxamate acts as a bidentate ligand.

Similarly, poly(N-vinyl lactamide) (−1807 kJ/mol, secondary hydroxyl α to amide) outperforms poly(N-vinyl-2-hydroxyacetamide) (−1576 kJ/mol, primary hydroxyl), consistent with known cooperativity of hydrogen bonds in constrained geometries.

4.6 Hydrazide as a high-performance motif

The final iterations introduced hydrazide derivatives, which proved highly effective. Poly(N-acetyl-N-methyl-hydrazinoacetamide) (−2145 kJ/mol, iteration 25) entered the all-time top 3. The secondary hydrazide (N-methyl substitution on hydrazine) combined with a secondary amide provides a hydrogen-bonding geometry analogous to the parent C2 scaffold with the additional stabilization of the hydrazide NH donor. The primary hydrazide variant scored −1698 kJ/mol, confirming that secondary hydrazide geometry is superior—mirroring the primary/secondary amide trend observed in the top scaffold.

4.7 Comparison of search strategies' chemical coverage

The three non-agentic methods explore fundamentally different regions of chemical space. RL converges to 14 unique PSMILES dominated by esters, ethers, and simple carbonyls (best: −1412 kJ/mol)—functional groups accessible through single-step mutations from the PEG seed. Optuna discovers 26 unique candidates including the urea bridge [*]NC(=O)NC([*])=O (−1683 kJ/mol) and peroxide-ester [*]OOC(=O)C([*])=O (−1171 kJ/mol), the latter illustrating the risk of energy-only optimization without chemical judgment (O–O peroxide bonds are thermally labile).

The agentic workflow produces 25+ unique candidates spanning polysaccharides (chitosan, iteration 1), acrylates (iterations 3–7), heterocyclic urethanes (iterations 8–11), linear amides (iterations 12–16), and novel motifs including biuret, hydroxamic acid, and hydrazide scaffolds (iterations 17–25). This breadth arises from the agent's ability to propose entire polymer families based on literature context, rather than applying point mutations to a seed PSMILES. The agent's chemical reasoning—visible in the iteration logs—shows explicit hypothesis formation ("C2 backbone with dual amide is optimal because...") and targeted falsification (testing the C3 analog, the thio variant, and the ring-expanded analog to confirm the hypothesis).

5 Discussion

Agentic advantage is structural, not statistical. The 68% improvement of the agentic workflow over RL and 19% over Optuna does not arise from evaluating more candidates—the agentic campaign (115+ evaluations) is comparable to Optuna (143) and RL (∼100). Rather, the advantage is *structural*: the LLM proposes qualitatively different polymer families based on literature context (polysaccharides, heterocyclic urethanes, linear amides, hydrazides), while RL and Optuna are confined to single-step mutations from a seed PSMILES. The discovery of the C2 amide scaffold at iteration 15 required traversing a chemical space that no mutation operator in the RL library could reach from PEG.

Energy vs. synthesizability. The interaction energy metric captures non-bonded attraction in a near-static geometry but does not account for synthetic accessibility, thermal stability of the polymer, or kinetic aspects of insulin release. The peroxide-ester discovered by Optuna ([*]OOC(=O)C([*])=O, −1171 kJ/mol) illustrates the risk: a domain expert would immediately flag the labile O–O bond. The agentic workflow avoids such candidates through implicit chemical knowledge, but systematic integration of synthesizability scores (e.g. SAScore [11]) into the objective is planned.

Stochastic variability in screening. The replication of poly(N-acetyl-1,2-diaminoethane) in iteration 16 (−1909 kJ/mol vs. −2263 kJ/mol in iteration 15) reveals ∼15% variability in the Packmol-matrix protocol due to different random packing configurations. This noise level means that energy differences below ∼200 kJ/mol should be interpreted cautiously, while the ∼900 kJ/mol gap between the agentic best and RL is highly significant.

World model and coherence. The discovery world file addresses a key limitation of chat-based agentic systems: loss of context over long conversations. By persisting structured claims and hypotheses as a JSON artifact, the agent can "remember" across 25 iterations without relying on the LLM's context window. The agent's explicit hypothesis formation ("C2 backbone is optimal") and targeted falsification (C3 analog, thio variant, ring expansion) across iterations 14–21 demonstrate that the world model enables genuine scientific reasoning, not merely pattern matching. This is conceptually similar to the world model in Kosmos [16], adapted for a single-agent, human-in-the-loop setting.

Limitations of the screening protocol. The Packmol-matrix protocol packs polymer chains around insulin in vacuum, minimizes, and computes a single-point interaction energy. It does not sample conformational ensembles, account for explicit water, or compute binding free energies. We use it as a computationally tractable filter (seconds per candidate) rather than a predictive model. Planned extensions include NPT molecular dynamics with the UMA universal force field [19] and free-energy perturbation calculations for the top three candidates.

Comparison with Kosmos. Kosmos [16] executes ∼42,000 lines of code and reads 1,500 papers per 12-hour run, achieving 79.4% statement accuracy in generated reports. Our platform operates at a smaller scale (tens of candidates per iteration, ∼20 papers per mining call) but provides two capabilities Kosmos lacks: physics-grounded molecular screening through explicit simulation, and iterative structure–activity reasoning that produces falsifiable hypotheses and quantitative design rules.

6 Conclusions

We have presented an agentic materials discovery platform for insulin delivery polymers that integrates LLM-driven literature mining, cheminformatics mutation, and OpenMM-based molecular screening through a Model Context Protocol server with a structured discovery world model.

Benchmarking four search strategies reveals a clear hierarchy: RL (DQN, PPO) provides fast search with best energies near -1345 kJ/mol but limited chemical diversity (14 unique PSMILES); Optuna TPE achieves -1902 kJ/mol through broader exploration at $15\times$ the wall time; and the LLM-orchestrated agentic workflow reaches -2263 kJ/mol—a 19% improvement over Optuna and 68% over RL—by leveraging literature context and iterative structure–activity reasoning to discover polymer families inaccessible to mutation-based optimizers.

The agentic campaign identifies three breakthrough scaffolds for insulin stabilization, all sharing a C2 backbone with multiple amide groups. Structure–activity analysis across 25 iterations establishes quantitative design rules: the C2 backbone geometry is critical (C3 fails to pack); primary amine combined with secondary and primary amide groups provides optimal hydrogen-bond complementarity with insulin; ring expansion, thio-substitution, and bulky N-substituents systematically degrade performance; and hydroxamic acid and hydrazide motifs provide high-performance alternatives to the parent amide scaffold.

These results demonstrate that LLM-orchestrated, physics-grounded search produces both superior candidates and chemically interpretable design rules compared to classical optimization methods. The top three candidates—poly(N-acetyl-1,2-diaminoethane), poly(N-hydroxyacetyl-1,2-diaminoethane), and poly(N-acetyl-N-methyl-hydrazinoacetamide)—merit experimental validation for insulin stabilization at elevated temperatures and transdermal patch fabrication. Code, benchmark scripts, and full iteration logs are available in the project repository.

7 Acknowledgments

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